

1 **Reinforcing kōura resilience using rock piles to**
2 **enhance freshwater crayfish habitat in a**
3 **Aotearoa New Zealand lake**

4 **Stone piles Rotoiti**

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8 Placeholder — abstract to be written.

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1 Introduction

Freshwater biodiversity is declining faster than any other biome, with one quarter of freshwater fauna now threatened with extinction (Sayer *et al.*, 2025). Among the most endangered are freshwater crayfish, with 32% of species assessed as threatened globally due to habitat loss, invasive species, and water quality decline (Richman *et al.*, 2015). As keystone species and ecosystem engineers, crayfish play crucial roles in nutrient cycling, bioturbation, and foodweb dynamics, and their loss can have cascading consequences for freshwater ecosystems (Momot, 1995; Collier, Parkyn & Rabeni, 1997; Parkyn, Collier & Hicks, 2001).

In Aotearoa New Zealand (hereafter Aotearoa), the endemic freshwater crayfish kōura (*Paranephrops planifrons* White, 1842) is a taonga (culturally treasured) species of significant ecological and cultural importance to Māori, particularly for Te Arawa iwi of the Rotorua lakes region in the North Island of Aotearoa (Kusabs & Quinn, 2009). Kōura support customary fisheries using cultural practices such as tau kōura, yet populations in Lakes Rotoiti and Rotorua have declined by up to 96 % over the past two decades, with consequent loss of these customary practices (Kusabs *et al.*, 2015a; Kusabs *et al.*, 2026). The primary driver of decline is predation by the invasive brown bullhead catfish (*Ameiurus nebulosus* Lesueur, 1819, hereafter catfish), confirmed in these lakes in 2016 and 2018 (Francis, 2019; Grayling, 2020). Catfish are native to eastern North America where they mostly live in muddy slow-moving water bodies and can survive and spawn in a wide temperature range, tolerate poor water quality, low oxygen concentration and have a broad diet (Scott & Crossman, 1973), making them a highly adaptable/tolerant invader. Catfish were introduced in Aotearoa in 1877 as sport fish (McDowall, 1994) with their distribution continuously spreading over the upper North Island such as the Waikato River and in 1985 they were detected in Lake Taupō (Barnes & Hicks, 2003). In Aotearoa catfish are a major predator on kōura, as 64 % of large catfish (<250 mm fork length) had kōura in their stomach contents (Barnes & Hicks, 2003). In combination with predation by catfish kōura are further compressed by multiple interacting stressors that progressively restrict their available habitat. Kōura are naturally found in complex rocky habitats (Jowett, Parkyn & Richardson, 2008; Kusabs, Quinn & Hamilton, 2015b) or in deeper parts of the lake to avoid predation during the day, migrating to the productive littoral zone at night to forage (Devcich, 1979). However, thermal stratification creates anoxic conditions in deeper waters, displacing kōura into the shallow littoral zone

(Kusabs *et al.*, 2026), precisely where catfish predation pressure is greatest, as catfish rarely swim beyond 17 m depth (Dedual, 2002). This compression is further intensified by dense invasive macrophyte beds obstructing kōura migration in the littoral zone (Kusabs & Quinn, 2009), which also intensify oxygen depletion in deeper waters by die-off events (Vincent, Gibbs & Dryden, 1984; Hamilton, McBride & Uraoka, 2005). The importance of deep refuge is shown by Lake Taupō, where kōura persist at a depth even in the presence of catfish because the lake remains oxygenated below the thermocline (Verburg & Albert, 2019). In Lake Rotoiti this deep refuge becomes anoxic and unavailable for kōura (Kusabs *et al.*, 2026). Next to direct predation catfish exposure triggers anti-predator responses in kōura including increased refuge use (Raven *et al.*, (a) in review), reducing time available for foraging and potentially suppressing growth and reproduction (Preisser, Bolnick & Benard, 2005; Sheriff *et al.*, 2020). Climate change is expected to strengthen thermal stratification, expand anoxic zones (Woolway *et al.*, 2021), and increase the range of suitable catfish habitat in Aotearoa, further reducing kōura persistence (Lee *et al.*, 2025).

The compression of kōura into the littoral zone makes the quality of nearshore habitats increasingly important for kōura populations. In stream environments, kōura are positively associated with structurally complex habitats including bank undercuts, woody debris, leaf litter, and coarse substrates (Usio & Townsend, 2000; Parkyn & Collier, 2004; Jowett *et al.*, 2008; Parkyn, Meleason & Davies-Colley, 2009). In lake environments benthic substrate characteristics are a stronger predictor of kōura distribution than nutrient enrichment or predator abundance (Kusabs *et al.*, 2015b), and kōura are positively associated with coarse substrates and riparian vegetation (Raven *et al.*, (b) in review). Coarse rocky substrate provides refugia from predators and shelter during moulting (Streissl & Hödl, 2002). Experimental studies confirm that shelter availability directly reduces catfish predation on crayfish (Adams, 2007; Zhao *et al.*, 2024), and kōura increase refuge use in the presence of catfish compared to native predators (Raven *et al.*, (a) in review), emphasising the importance of shelter availability for kōura populations. Shelter availability is likely a limiting factor for kōura in Lake Rotoiti, where the majority of the shoreline is sandy with dense beds of introduced macrophytes and little natural rocky structure. Adding rocky substrate to soft-bottomed lake areas has been shown to significantly increase crayfish abundance (Johnsen & Taugbøl, 2008), and increasing structural complexity in streams improves crayfish restocking success (Huolila, Marjomäki & Laukkanen, 1997). Deploying habitat enhancement structures that add structural complexity in the littoral zone could potentially be a practical way to restore kōura populations in Lake Rotoiti.

18 In this study habitat enhancement structures in the shape of stone piles were deployed along
19 the shoreline of Lake Rotoiti to investigate the ability of the structures in providing additional
20 habitat for kōura to live and hide from predators. Kōura presence, abundance, and biomass
21 were monitored at stone pile, control, and reference sites over 18 months. We predicted that
22 kōura occupancy, abundance and biomass would increase at stone pile sites relative to control
23 sites after construction, and that catfish would not change. We further predicted that kōura
24 abundance would keep increasing progressively after construction, at least within the first 18
25 months, and that catfish would instead show seasonal patterns of shoreline use rather than a
26 construction-driven trend (Barnes, 1996; Dedual, 2002). Finally, we predicted that stone pile
27 sites would progressively converge toward kōura densities observed at natural rocky reference
28 sites, and that catfish would not show this pattern.

29 **2 Materials and Methods**

1 **2.1 Study location and construction**

2 Lake Rotoiti (-38.03, 176.42) is a warm monomictic mesotrophic lake located in the Rotorua Te
3 Arawa Lakes region of the North Island of Aotearoa (34 km², maximum depth 128 m). The lake
4 stratifies for approximately nine months a year. Since 2008, nutrient-rich inflows from Lake
5 Rotorua have been diverted toward the Kaituna River via the Ōhau Channel diversion wall to
6 improve water quality in Lake Rotoiti (Hamilton *et al.*, 2005; Prentice & Özkundakci, 2024).
7 Lake level is regulated by a weir at the outlet of the Kaituna River, providing a stable and
8 controlled water level throughout the year. Catfish are present throughout the lake but occur
9 in particularly high concentrations in Te Weta Bay on the North West side of the lake (Francis,
10 2019). Five stone pile sites were selected alongside the lake shoreline based on accessibility and
11 land owner approval (Figure 1). At each stone pile site, a paired control site was established
12 50 m along the shoreline in the same habitat type. Five naturally occurring rocky shoreline
13 locations were selected as reference sites, each with two monitoring points 50 m apart. In
14 total 20 sites of 150 m² (15 * 10 m) were monitored seasonally over 18 months. Baseline
15 monitoring was conducted in February 2025 prior to stone pile deployment, which occurred
16 the following week. Stone piles were constructed from locally sourced quarry rock (Manawahe
17 Stone and Quarry, ~25 km from the lake) with a size range of 50 - 400 mm. At each of the
18 five stone pile sites, three stone piles were arranged horizontally along the shoreline by hand
19 from the shore. Stone pile design was informed by mesocosm experiments demonstrating kōura

20 preference for enclosed stone structures (Raven *et al.*, (c) in review). Stone piles were positioned
21 approximately 4.3 m from shore at a mean depth of 0.49 m and with a mean area of 4.1 m².

22 2.2 Monitoring

23 Monitoring consisted of physical, chemical, and biological parameters recorded at each site
24 during each monitoring round. Static site characteristics including substrate composition, slope
25 to depth contour, riparian vegetation, overhanging tree cover and wood cover were visually
26 estimated at baseline monitoring only, as these were not expected to change substantially over
27 the monitoring period. Substrate composition was visually estimated using an underwater
28 viewing chamber, categorising bedrock, boulders (>256 mm), cobble (64–256 mm), gravel (2–64
29 mm), sand (0.063–2 mm), mud (<0.063 mm), and organic matter. A substrate index was
30 calculated from these estimates (Jowett & Richardson, 1990; Jowett *et al.*, 2008):

$$\text{Substrate index} = 0.08 \times \text{bedrock} + 0.07 \times \text{boulders} + 0.06 \times \text{cobble} + 0.04 \times \text{gravel} + 0.03 \times \text{sand} + 0.02 \times \text{organic matter}$$

31 Coarser substrates were assigned higher weights as they provide greater interstitial pore space
32 and stable structures used by kōura as refuge (Usio & Townsend, 2000; NYSTRÖM *et al.*,
1 2006). After stone pile construction, measurements of the size and dimensions of each structure
2 were taken. The post-construction substrate index for stone pile sites was recalculated using
3 each site's combined stone-pile footprint area, expressed as a percentage of the 150 m² site area
4 and treated as 80 % cobble and 20 % boulders displacing an equivalent area of sand, to reflect
5 their contribution to substrate complexity. Slope was assessed as the horizontal distance from
6 the shoreline to the 20 m depth contour, derived from bathymetric maps and measured in a
7 projected coordinate system (NZTM2000; EPSG:2193).

8 Seasonal variable parameters were recorded at every monitoring round, and included aquatic
9 vegetation cover (emergent and submerged macrophytes) and chemical parameters included
10 water temperature (°C), dissolved oxygen (mg L⁻¹), and conductivity (µS cm⁻¹) were measured
11 using a YSI ProSolo meter (Pro2030, YSI Inc., Yellow Springs, OH, USA). pH was measured
12 using a handheld pH meter (EC-PCTestr35, Eutech Instruments).

13 Kōura and fish were every monitoring round sampled using two single-winged, double-throated
14 fyke nets (600 mm high D-shaped entrance hoop) and two collapsible bait traps (480 × 250 mm;
15 two 50 mm entrance openings) baited with catfish flesh (MacNeil *et al.*, 2025). Fyke nets were
16 placed perpendicular to the shoreline outside of the stone pile structures or at 10 m intervals,

with bait traps positioned between them. All equipment was deployed simultaneously at each site for a 24-h period. Captured kōura were measured for orbital carapace length (OCL) using a digital calliper (0.01 mm precision; 150 mm) and for wet mass using a digital scale (± 1 g; 5 kg capacity, Anko Brand, Kmart Group, Mulgrave, VIC, Australia). Catfish and other fish species, including eels (*Anguilla* spp.), goldfish (*Carassius auratus* (Linnaeus, 1758)), common smelt (*Retropinna retropinna* (Richardson, 1848)), kōaro (*Galaxias brevipinnis* Günther, 1866), rainbow trout (*Oncorhynchus mykiss* Walbaum, 1792), common bully (*Gobiomorphus cotidianus* McDowall & Fulton, 1978), and mosquitofish (*Gambusia affinis* (S. F. Baird & Girard, 1853)), were recorded for abundance and measured for total length, and weighed for wet mass. Catch per unit effort (CPUE) and biomass per unit effort (BPUE) were calculated for all kōura and fish based on the number of nets and traps used at each site for each monitoring round (Kusabs *et al.*, 2015b). Missing kōura weights due to failure of scale ($n = 36$) were estimated using a log –log length–weight regression fitted to individuals with both measurements ($n = 757$), following standard approaches for weight length relationships in aquatic organisms Figure 5. Predicted values were back-transformed with a lognormal bias correction factor ($10^{\{0.5 \cdot \hat{\sigma}^2\}}$) (Quinn & Keough, 2002).

$$\log_{10}(\text{Weight [g]}) = -2.67 + 2.683 \times \log_{10}(\text{OCL length [mm]})$$

2.3 Data analysis

Site variation before stone pile construction was assessed using paired Wilcoxon signed-rank tests to compare stone pile sites with their paired control sites, and unpaired Wilcoxon rank-sum tests to compare stone pile sites with reference sites. This was done for all response variables and site characteristics to confirm that the stone pile and control sites were equivalent before construction, and to characterise the difference between stone pile and reference sites. We fitted four complementary generalised linear mixed effects models (GLMMs) to assess the effect of construction: a construction-effect model, testing whether kōura or catfish occupancy, abundance, and biomass responded to stone pile construction relative to control sites in a before-after, control-impact (BACI) design (Underwood, 1994); a post-construction trend model, testing whether that response changed progressively over time at stone pile relative to control sites; a convergence model, testing whether stone pile sites approached reference-sites densities over time; and a baseline and equivalence model, testing whether treatment and reference sites were equivalent before construction and after construction.

20 The construction-effect model combined all six post-construction monitoring rounds into one
 21 factor level, retaining each round as a separate observation. The occupancy response variable
 22 was fitted with a Binomial family with logit link and the abundance and biomass with a Tweedie
 23 familie loglink. In the model, period (before/after) and site were fitted as an interaction, with
 24 monitoring number and temperature as covariates and site nested within pair as a random
 1 effect.

2 The post-construction trend model tested the response variables kōura catfish abundance or
 3 biomass fitted with Tweedie, site type and monitoring number as interaction, temperature and
 4 substrate index from before construction as covariates and site number as random effect.

5 The convergence model tested the response variables kōura catfish abundance or biomass fitted
 6 with Tweedie, site type and monitoring number as interaction, season and substrate index from
 7 post-construction as covariates and site number as random effect.

8 The baseline and equivalent models tested the response variables kōura catfish abundance or
 9 biomass fitted with Tweedie, site type as fixed effect for the baseline and for the equivalente
 10 models the site type and monitoring number as interaction and temperature as covariate and
 11 site nested within pair as a random effect.

12 All models were fitted and validated in R (version 4.5.2; R Core Team (2025)) using the
 13 glmmTMB, DHARMA, and emmeans packages (Hartig, 2016; Brooks *et al.*, 2017; Lenth
 14 & Piaskowski, 2017). Multiple-testing correction using Benjamini–Hochberg correction was
 15 applied separately for each model. Robustness of the models was further assessed via sensitivity
 16 analyses and colonisation driver comparison in supplementary material.

17 **3 Results**

18 Construction-effect model

19 Post-construction trend model

20 Convergence model

21 Figure 2

22 Figure 3

1 Figure 4

2 4 Discussion

3 Most kōura were found along the edges of the stone piles there for recommended future stone
4 piles should be designed to create the most edges.

5 Stone piles were constructed at the shore line to shallow to reach macrophyte beds which
6 are located further offshore. Future stone pile construction could be extended further offshore
7 to reach the macrophyte beds and test if stone piles could suppress macrophyte growth and
8 create pathways for kōura to move between the deeperparts of the lake and the shore.

9 4.1 Conclusion

10 5 Acknowledgements

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4 Council Chair in Lake and Freshwater Science programme.

5 5.2 Authors’ contributions

6 Study conception and design were undertaken by O.V.R., I.A.K.K., R.H., F.J.B. and D.Ö.
7 Fieldwork was executed by O.V.R. Data analysis was performed by O.V.R. I.A.K.K obtained
8 council and iwi consent and provided local ecological knowledge. Manuscript drafting was led

9 by O.V.R. All authors contributed to manuscript revision, approved the results, and the final
1 manuscript.

2 **5.3 Use of Artificial Intelligence**

3 Use of Artificial Intelligence: Claude (Anthropic) was used to assist with manuscript editing and
4 language refinement. All content was critically reviewed and the authors take full responsibility
5 for the final manuscript.

6 **5.4 Declarations**

7 The corresponding author has declared that none of the authors has any competing interests.

8 **5.5 Data availability statement**

9 The code and derived data supporting the findings of this study are openly available on GitHub
10 (https://github.com/OlivierRaven/Stone_Piles_Rotoiti) and a rendered version of the analysis
11 is hosted at https://olivierraven.github.io/Stone_Piles_Rotoiti/. The primary raw dataset is
12 archived on Zenodo (<https://doi.org/.....>). Peer review documents, including the cover letter
13 and reviewer responses, are available in the manuscript repository in the interest of open and
14 transparent science.

15 **6 Tables**

Table 1: Baseline (pre-construction) comparability of stone pile, control, and reference sites. Values are mean $\pm$ SD; p-values test Stone pile vs Control (paired Wilcoxon signed-rank test, pairs matched by pair_id, supporting the BACI design) and Stone pile vs Reference (unpaired Wilcoxon rank-sum test, documenting the starting gap relative to natural conditions, motivating the convergence analysis). With only 5 paired sites, these are low-power checks — a non-significant result indicates no obvious gross imbalance, not formal proof of equivalence.

Variable	Stone_pile	Control	Reference	p_vs_Con- trol	p_vs_Ref- erence
Kōura CPUE	0.2 $\pm$ 0.27	0.15 $\pm$ 0.22	2.13 $\pm$ 2.32	1.000	0.205
Catfish CPUE	0.05 $\pm$ 0.11	0.1 $\pm$ 0.22	0.3 $\pm$ 0.52	1.000	0.420
Kōura BPUE	7.95 $\pm$ 12.15	2.8 $\pm$ 3.84	36.69 $\pm$ 36.82	0.789	0.206
Catfish BPUE	0.15 $\pm$ 0.34	24.6 $\pm$ 55.01	97.28 $\pm$ 232.15	1.000	0.343
Substrate index (before)	3.39 $\pm$ 1.07	3.68 $\pm$ 0.95	5.29 $\pm$ 1.05	0.789	0.012
Substrate index (after)	3.65 $\pm$ 1.12	3.68 $\pm$ 0.95	5.29 $\pm$ 1.05	1.000	0.032
Slope (0–20 m)	0.06 $\pm$ 0.05	0.06 $\pm$ 0.04	0.09 $\pm$ 0.05	0.590	0.245
Temperature (°C)	24.74 $\pm$ 1.92	24.78 $\pm$ 1.47	24.48 $\pm$ 0.78	1.000	0.623
Dissolved oxygen (mg/L)	9.85 $\pm$ 1.48	9.63 $\pm$ 1.29	10.03 $\pm$ 0.86	0.418	0.426
Specific conductivity (μ S/cm)	195.44 $\pm$ 70.05	199.8 $\pm$ 78.52	165.45 $\pm$ 8.5	0.281	0.951
pH	7.5 $\pm$ 2.15	7.61 $\pm$ 2.2	8.53 $\pm$ 0.66	1.000	0.426

¹⁶ Source: [Article Notebook](#)

¹⁷ 7 Figures

¹⁸ 8 Supplementary information



Figure 1: Map of the monitoring sites.

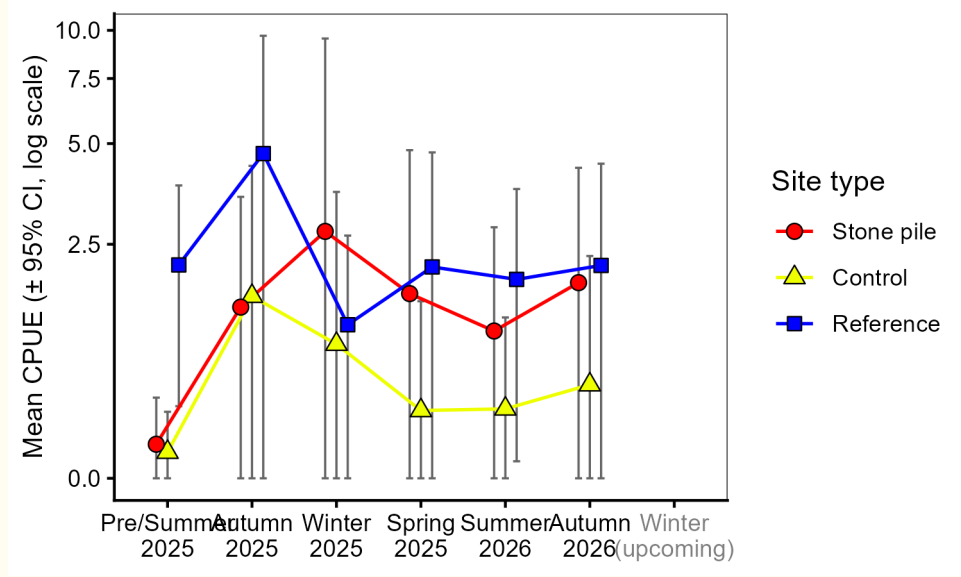


Figure 2: Mean catch per monitoring round ($\pm$ 95% CI) by site type for kōura.

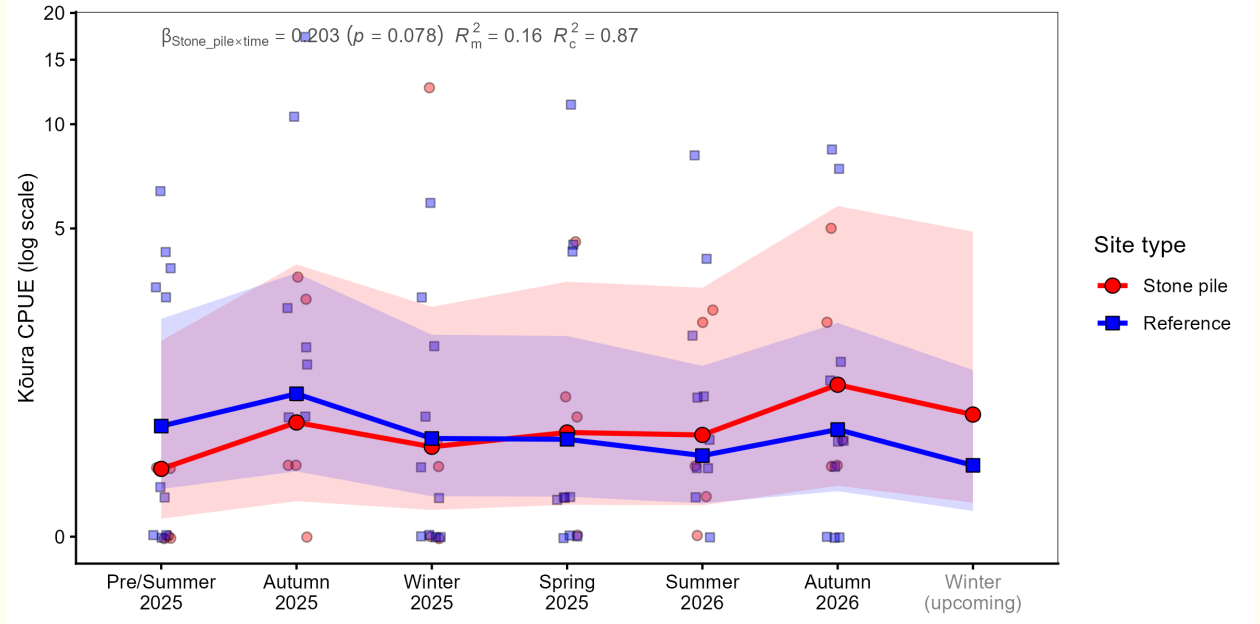


Figure 3: Predicted kōura CPUE trajectories over monitoring rounds at stone pile and reference sites from the RQ3 abundance convergence model. Shaded ribbons represent 95% confidence intervals. The annotation shows the key interaction term ($\pm$ p-value) and model R^2 values for fixed effects (R^2_m) and conditional (R^2_c).

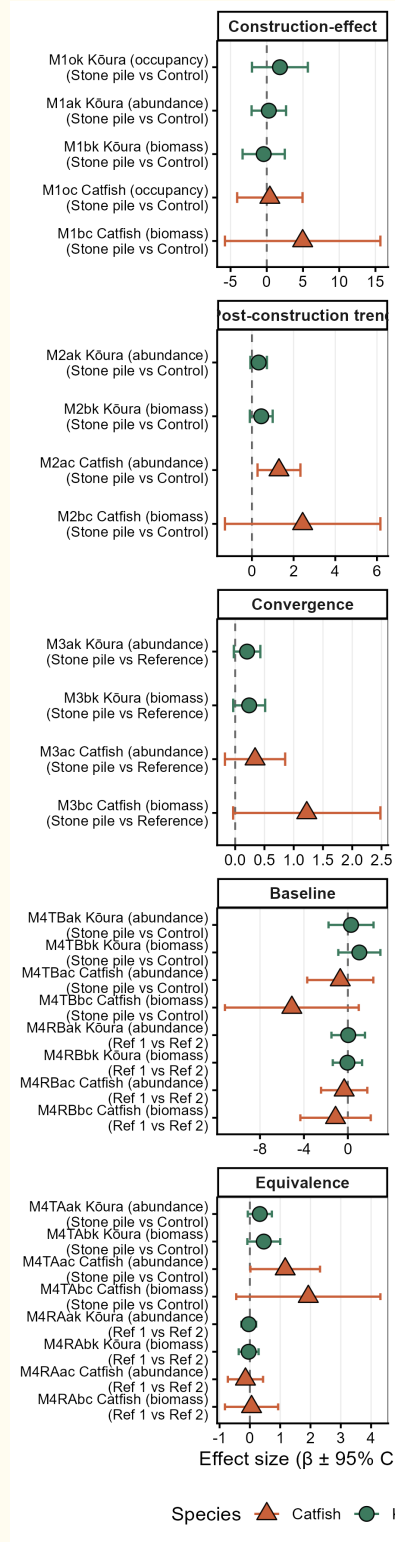


Figure 4: Effect sizes ($\beta \pm 95\% \text{ CI}$) for the key term of interest of every model in model_registry (kōura and catfish, RQ1–RQ4). Filled points indicate significant effects ($p < 0.05$). Each panel has its own x-axis scale— magnitudes are not directly comparable across panels because each RQ tests a different contrast (step-change, per-round slope, or simple group difference) and RQ1 occupancy is on the log-odds scale while all others use the Tweedie log link. Models with degenerate confidence intervals (non-positive-definite Hessian) are excluded.

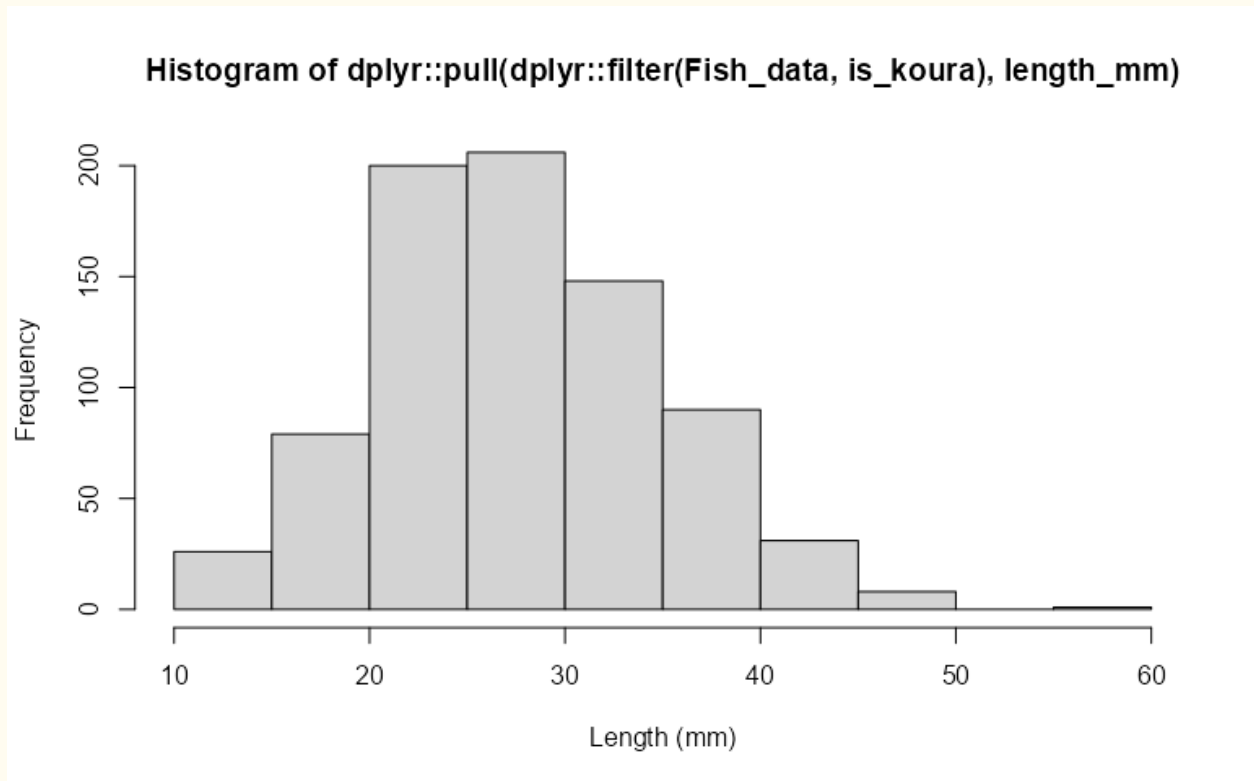


Figure 5: Length–weight relationship for kōura, fitted on $\log -\log$ scale. Open points are weights predicted from the length–weight regression (used to fill missing scale measurements); filled points are directly measured weights.

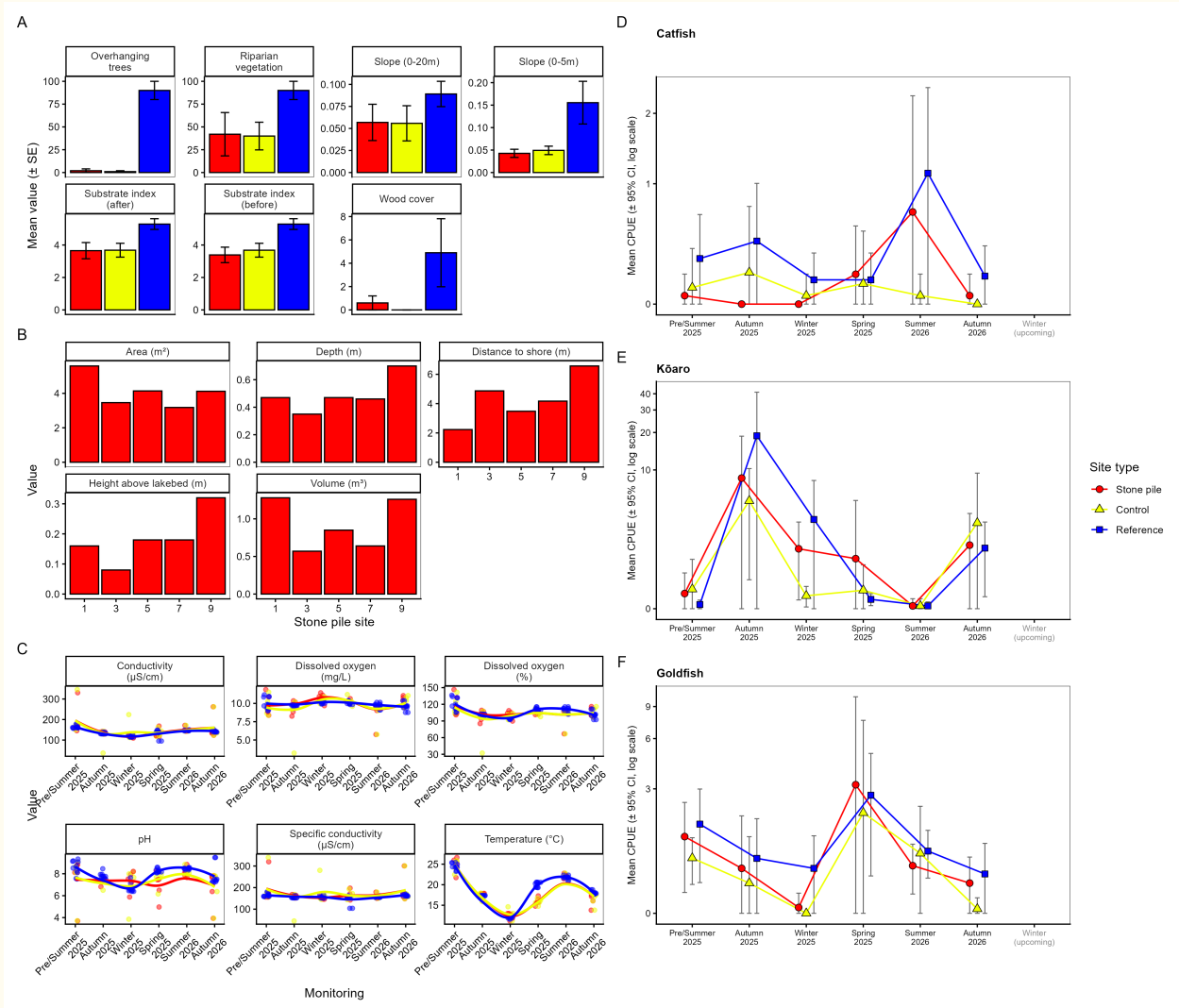


Figure 6: Descriptive context plots, combined. Left column (top to bottom): baseline habitat characteristics (mean $\pm$ SE by site type), stone pile structure dimensions (5 stone pile sites), chemical conditions over time (mean $\pm$ smoothed trend by site type). Right column: mean catch per monitoring round by site type, for catfish, kōaro, and goldfish. Site-type colours are shared across all panels — see the legend on the right. Background/methods-supporting figures, not primary results — see Results for the BACI, trend, and convergence models.

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	re-species	com- pari- response_type	time_frame	es- ti- mate	std.er-	statis- tic	p.value	adjco- BIC	low	high
M1okRQ1Kōura	occu- pancy	vs_confull	(Intercept)	- 4.736	-	0.090	NA	- 1.249		
		trol		8.033	1.696			17.315		
M1okRQ1Kōura	occu- pancy	vs_confull	PeriodAfter	4.171	2.500	1.669	0.095	NA	- 9.071	
		trol						0.728		
M1okRQ1Kōura	occu- pancy	vs_confull	siteStone_pile	0.017	2.482	0.007	0.995	NA	- 4.883	
		trol						4.849		
M1okRQ1Kōura	occu- pancy	vs_confull	monitoring_int	- 0.333	-	0.512	NA	- 0.435		
		trol		0.218	0.655			0.871		
M1okRQ1Kōura	occu- pancy	vs_confull	Temperature	0.292	0.171	1.713	0.087	NA	- 0.626	
		trol						0.042		
M1okRQ1Kōura	occu- pancy	vs_confull	Peri- odAfter:site- Stone_pile	1.814	1.968	0.922	0.357	0.853	- 5.671	
		trol						2.043		
M1akRQ1Kōura	abun- dance	vs_confull	(Intercept)	- 1.553	-	0.327	NA	- 1.523		
		trol		1.521	0.979			4.566		
M1akRQ1Kōura	abun- dance	vs_confull	PeriodAfter	1.659	1.043	1.591	0.112	NA	- 3.703	
		trol						0.384		
M1akRQ1Kōura	abun- dance	vs_confull	siteStone_pile	0.702	1.306	0.538	0.591	NA	- 3.261	
		trol						1.857		
M1akRQ1Kōura	abun- dance	vs_confull	monitoring_int	- 0.096	-	0.349	NA	- 0.099		
		trol		0.090	0.937			0.279		
M1akRQ1Kōura	abun- dance	vs_confull	Temperature	- 0.042	-	0.235	NA	- 0.032		
		trol		0.050	1.189			0.132		
M1akRQ1Kōura	abun- dance	vs_confull	Peri- odAfter:site- Stone_pile	0.283	1.214	0.233	0.816	0.853	- 2.661	
		trol						2.096		
M1bkRQ1Kōura	biomass	vs_confull	(Intercept)	0.883	1.951	0.452	0.651	NA	- 4.706	
		trol						2.941		

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	re-	com-	es-	ti-	std.er-	statis-	p.value	adjco	BLow	high
Q1	species	response_type	time_frame	mate	ror	tic	p.value	adjco	BLow	high
M1bkRQ1	Kōura	biomassvs_trol	PeriodAfter	1.786	1.374	1.300	0.194	NA	-	4.479
									0.907	
M1bkRQ1	Kōura	biomassvs_trol	siteStone_pile	1.370	1.550	0.884	0.377	NA	-	4.407
									1.667	
M1bkRQ1	Kōura	biomassvs_trol	monitoring_int	-	0.133	-	0.751	NA	-	0.219
				0.042		0.317			0.304	
M1bkRQ1	Kōura	biomassvs_trol	Temperature	-	0.058	-	0.708	NA	-	0.091
				0.022		0.374			0.135	
M1bkRQ1	Kōura	biomassvs_trol	Peri-	-	1.487	-	0.774	0.853	-	2.488
			odAfter:site-	0.426		0.287			3.341	
			Stone_pile							
M1locRQ1	Cat-	occu- vs_confull	(Intercept)	-	4.271	-	0.150	NA	-	2.225
	fish	pancy trol		6.146		1.439			14.517	
M1locRQ1	Cat-	occu- vs_confull	PeriodAfter	0.402	2.336	0.172	0.863	NA	-	4.980
	fish	pancy trol							4.176	
M1locRQ1	Cat-	occu- vs_confull	siteStone_pile	0.005	2.095	0.002	0.998	NA	-	4.110
	fish	pancy trol							4.101	
M1locRQ1	Cat-	occu- vs_confull	monitoring_int	0.112	0.343	0.325	0.745	NA	-	0.784
	fish	pancy trol							0.561	
M1locRQ1	Cat-	occu- vs_confull	Temperature	0.160	0.147	1.091	0.275	NA	-	0.449
	fish	pancy trol							0.128	
M1locRQ1	Cat-	occu- vs_confull	Peri-	0.426	2.302	0.185	0.853	0.853	-	4.939
	fish	pancy trol	odAfter:site-						4.087	
			Stone_pile							
M1lacRQ1	Cat-	abun- vs_confull	(Intercept)	-	NA	NA	NA	NA	NA	NA
	fish	dance trol		9.453						
M1lacRQ1	Cat-	abun- vs_confull	PeriodAfter	2.569	NA	NA	NA	NA	NA	NA
	fish	dance trol								

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model	Q1	species	re- response	com- pari- type	time_frame	es- ti- mate	std.er- ror	statis- tic	p.value	adjco- B	low	high
M1ac	RQ1	Cat- fish	abun- dance	vs_con- trol	full siteStone_pile	- 0.721	NA	NA	NA	NA	NA	NA
M1ac	RQ1	Cat- fish	abun- dance	vs_con- trol	full monitoring_int	- 0.422	NA	NA	NA	NA	NA	NA
M1ac	RQ1	Cat- fish	abun- dance	vs_con- trol	full Temperature	0.233	NA	NA	NA	NA	NA	NA
M1ac	RQ1	Cat- fish	abun- dance	vs_con- trol	full Peri- odAfter:site- Stone_pile	1.872	NA	NA	NA	NA	NA	NA
M1ac	RQ1	Cat- fish	abun- dance	vs_con- trol	full (Intercept)	- 1.472	NA	NA	NA	NA	NA	NA
M1bc	RQ1	Cat- fish	biomass	vs_con- trol	full (Intercept)	0.544	6.130	0.089	0.929	NA	- 11.470	12.558
M1bc	RQ1	Cat- fish	biomass	vs_con- trol	full PeriodAfter	- 0.414	3.271	- 0.127	0.899	NA	- 6.824	5.997
M1bc	RQ1	Cat- fish	biomass	vs_con- trol	full siteStone_pile	- 5.230	5.616	- 0.931	0.352	NA	- 16.238	5.778
M1bc	RQ1	Cat- fish	biomass	vs_con- trol	full monitoring_int	- 0.125	0.631	- 0.199	0.842	NA	- 1.362	1.111
M1bc	RQ1	Cat- fish	biomass	vs_con- trol	full Temperature	0.042	0.212	0.199	0.842	NA	- 0.373	0.457
M1bc	RQ1	Cat- fish	biomass	vs_con- trol	full Peri- odAfter:site- Stone_pile	4.954	5.473	0.905	0.365	0.853	- 5.774	15.682
M2ak	RQ2	Kōura- abun- dance	vs_con- post	control	(Intercept)	- 2.259	2.418	- 0.934	0.350	NA	- 6.998	2.480
M2ak	RQ2	Kōura- abun- dance	vs_con- post	control	siteStone_pile	0.457	1.196	0.382	0.702	NA	- 1.886	2.800

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model	re-	com-	es-	ti-	std.er-	stat-	p.value	adjco-	BH	low	high		
Q1	species	response	type	time	frame	mate	ror	tic	p.value	adjco-	BH	low	high
M2akRQ2	Kōura	abun-	vs_conpost	monitoring_int		-	0.164	-	0.074	NA	-	0.028	
	dance	trol				0.294		1.790		0.616			
M2akRQ2	Kōura	abun-	vs_conpost	Temperature		-	0.043	-	0.205	NA	-	0.030	
	dance	trol				0.055		1.267		0.140			
M2akRQ2	Kōura	abun-	vs_conpost	Substrate_in-		0.785	0.563	1.395	0.163	NA	-	1.888	
	dance	trol		dex_before						0.318			
M2akRQ2	Kōura	abun-	vs_conpost	site-		0.322	0.202	1.593	0.111	0.148	-	0.718	
	dance	trol		Stone_pile:mon-						0.074			
				itoring_int									
M2bkRQ2	Kōura	biomass	vs_conpost	(Intercept)		1.128	2.305	0.489	0.625	NA	-	5.645	
		trol								3.389			
M2bkRQ2	Kōura	biomass	vs_conpost	siteStone_pile		-	1.215	-	0.991	NA	-	2.367	
		trol				0.014		0.012		2.396			
M2bkRQ2	Kōura	biomass	vs_conpost	monitoring_int		-	0.228	-	0.157	NA	-	0.124	
		trol				0.323		1.417		0.769			
M2bkRQ2	Kōura	biomass	vs_conpost	Temperature		-	0.060	-	0.685	NA	-	0.093	
		trol				0.024		0.405		0.142			
M2bkRQ2	Kōura	biomass	vs_conpost	Substrate_in-		0.607	0.494	1.230	0.219	NA	-	1.576	
		trol		dex_before						0.361			
M2bkRQ2	Kōura	biomass	vs_conpost	site-		0.448	0.280	1.601	0.109	0.148	-	0.996	
		trol		Stone_pile:mon-						0.100			
				itoring_int									
M2acRQ2	Cat-	abun-	vs_conpost	(Intercept)		-	4.312	-	0.643	NA	-	6.456	
	fish	dance	trol			1.996		0.463		10.448			
M2acRQ2	Cat-	abun-	vs_conpost	siteStone_pile		-	2.402	-	0.123	NA	-	1.003	
	fish	dance	trol			3.705		1.542		8.413			
M2acRQ2	Cat-	abun-	vs_conpost	monitoring_int		-	0.385	-	0.029	NA	-	-	
	fish	dance	trol			0.843		2.188		1.598	0.088		

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M2ac	RQ2	Cat-	abun-	vs_con	post	Temperature	0.244	0.085	2.875	0.004	NA	0.078	0.410
		fish	dance	trol									
M2ac	RQ2	Cat-	abun-	vs_con	post	Substrate_in-	-	1.051	-	0.348	NA	-	1.074
		fish	dance	trol		dex_before	0.986		0.938			3.046	
M2ac	RQ2	Cat-	abun-	vs_con	post	site-	1.298	0.525	2.471	0.013	0.054	0.269	2.328
		fish	dance	trol		Stone_pile:mon-							
						itoring_int							
M2ac	RQ2	Cat-	abun-	vs_con	post	(Intercept)	-	23564.174	-	0.999	NA	-	46163.743
		fish	dance	trol			21.188		0.001			46206.120	
M2bc	RQ2	Cat-	biomass	vs_con	post	(Intercept)	1.775	9.109	0.195	0.846	NA	-	19.629
		fish		trol								16.080	
M2bc	RQ2	Cat-	biomass	vs_con	post	siteStone_pile	-	8.189	-	0.298	NA	-	7.529
		fish		trol			8.521		1.041			24.570	
M2bc	RQ2	Cat-	biomass	vs_con	post	monitoring_int	-	0.888	-	0.232	NA	-	0.678
		fish		trol			1.062		1.196			2.802	
M2bc	RQ2	Cat-	biomass	vs_con	post	Temperature	0.111	0.228	0.485	0.628	NA	-	0.557
		fish		trol								0.336	
M2bc	RQ2	Cat-	biomass	vs_con	post	Substrate_in-	-	2.053	-	0.929	NA	-	3.841
		fish		trol		dex_before	0.183		0.089			4.206	
M2bc	RQ2	Cat-	biomass	vs_con	post	site-	2.434	1.903	1.279	0.201	0.201	-	6.164
		fish		trol		Stone_pile:mon-						1.295	
						itoring_int							
M3ak	RQ3	Kōura	abun-	vs_ref	full	(Intercept)	-	2.338	-	0.224	NA	-	1.742
			dance	er-			2.840		1.215			7.421	
				ence									
M3ak	RQ3	Kōura	abun-	vs_ref	full	siteStone_pile	0.393	1.175	0.335	0.738	NA	-	2.695
			dance	er-								1.909	
				ence									

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model	Q1	Q3	species	response	type	com- pari- time_frame	es- ti- mate	std.er- ror	statis- tic	p.value	adjco- B	How	low	high
M3akRQ3	Kōura	abun-	vs_ref-	dance	er-	monitoring_int	-	0.060	-	0.093	NA	-	0.017	
		dance			er-		0.101		1.680			0.219		
					ence									
M3akRQ3	Kōura	abun-	vs_ref-	dance	er-	SeasonSpring	-	0.262	-	0.210	NA	-	0.185	
		dance			er-		0.328		1.255			0.842		
					ence									
M3akRQ3	Kōura	abun-	vs_ref-	dance	er-	SeasonSummer	-	0.214	-	0.031	NA	-	-	
		dance			er-		0.463		2.162			0.882	0.043	
					ence									
M3akRQ3	Kōura	abun-	vs_ref-	dance	er-	SeasonWinter	-	0.264	-	0.113	NA	-	0.099	
		dance			er-		0.419		1.587			0.937		
					ence									
M3akRQ3	Kōura	abun-	vs_ref-	dance	er-	Substrate_in-	0.626	0.420	1.489	0.137	NA	-	1.449	
		dance			er-	dex_after						0.198		
					ence									
M3akRQ3	Kōura	abun-	vs_ref-	dance	er-	site-	0.203	0.115	1.761	0.078	0.114	-	0.430	
		dance			er-	Stone_pile:mon-						0.023		
					ence	itoring_int								
M3bkRQ3	Kōura	biomass	vs_ref-			(Intercept)	1.427	2.125	0.671	0.502	NA	-	5.592	
					er-							2.738		
					ence									
M3bkRQ3	Kōura	biomass	vs_ref-			siteStone_pile	-	1.091	-	0.923	NA	-	2.033	
					er-		0.105		0.096			2.243		
					ence									
M3bkRQ3	Kōura	biomass	vs_ref-			monitoring_int	-	0.075	-	0.249	NA	-	0.061	
					er-		0.087		1.153			0.234		
					ence									

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model	re- Q1	com- pari- response_type	time_frame	es- ti- mate	std.er- ror	statis- tic	p.value	adjco- B	low	high
M3bkRQ3	Kōura	biomass	vs_ref-full	SeasonSpring	- 0.342	- 0.127	NA	- 0.148		
		er- ence		0.522	1.528			1.193		
M3bkRQ3	Kōura	biomass	vs_ref-full	SeasonSummer	- 0.262	- 0.109	NA	- 0.094		
		er- ence		0.420	1.603			0.934		
M3bkRQ3	Kōura	biomass	vs_ref-full	SeasonWinter	- 0.334	- 0.153	NA	- 0.177		
		er- ence		0.477	1.430			1.132		
M3bkRQ3	Kōura	biomass	vs_ref-full	Substrate_in- dex_after	0.406	0.380	1.067	0.286	NA	- 1.150
		er- ence						0.339		
M3bkRQ3	Kōura	biomass	vs_ref-full	site- Stone_pile:mon- itoring_int	0.239	0.139	1.718	0.086	0.114	- 0.512
		er- ence						0.034		
M3acRQ3	Cat- fish	abun- dance	vs_ref-full	(Intercept)	- 3.698	- 0.124	NA	- 1.552		
		er- ence		5.696	1.540			12.944		
M3acRQ3	Cat- fish	abun- dance	vs_ref-full	siteStone_pile	- 2.125	- 0.615	NA	- 3.098		
		er- ence		1.067	0.502			5.233		
M3acRQ3	Cat- fish	abun- dance	vs_ref-full	monitoring_int	0.159	0.091	1.747	0.081	NA	- 0.338
		er- ence						0.019		
M3acRQ3	Cat- fish	abun- dance	vs_ref-full	SeasonSpring	- 0.547	- 0.852	NA	- 0.970		
		er- ence		0.102	0.187			1.174		

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M3ac	RQ3	Cat-	abun-	vs_ref-full	SeasonSummer	1.321	0.359	3.685	0.000	NA	0.618	2.024	
		fish	dance	er- ence									
M3ac	RQ3	Cat-	abun-	vs_ref-full	SeasonWinter	-	0.649	-	0.534	NA	-	0.869	
		fish	dance	er- ence		0.404		0.622			1.676		
M3ac	RQ3	Cat-	abun-	vs_ref-full	Substrate_in-	0.486	0.662	0.734	0.463	NA	-	1.785	
		fish	dance	er- ence	dex_after						0.812		
M3ac	RQ3	Cat-	abun-	vs_ref-full	site-	0.340	0.263	1.293	0.196	0.196	-	0.854	
		fish	dance	er- ence	Stone_pile:mon-						0.175		
					itoring_int								
M3ac	RQ3	Cat-	abun-	vs_ref-full	(Intercept)	-	10860.696	-	0.998	NA	-	21265.816	
		fish	dance	er- ence		20.757		0.002			21307.331		
M3bc	RQ3	Cat-	biomass	vs_ref-full	(Intercept)	-	5.884	-	0.789	NA	-	9.959	
		fish		er- ence		1.573		0.267			13.106		
M3bc	RQ3	Cat-	biomass	vs_ref-full	siteStone_pile	-	4.013	-	0.163	NA	-	2.270	
		fish		er- ence		5.596		1.394			13.462		
M3bc	RQ3	Cat-	biomass	vs_ref-full	monitoring_int	-	0.178	-	0.957	NA	-	0.339	
		fish		er- ence		0.010		0.054			0.358		
M3bc	RQ3	Cat-	biomass	vs_ref-full	SeasonSpring	-	0.919	-	0.608	NA	-	1.329	
		fish		er- ence		0.472		0.513			2.273		

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M3bcRQ3	Cat-	fish	biomass	svs_ref- er- ence	full SeasonSummer	0.878	0.664	1.323	0.186	NA	-	2.178
											0.423	
M3bcRQ3	Cat-	fish	biomass	svs_ref- er- ence	full SeasonWinter	0.207	0.861	0.240	0.810	NA	-	1.894
											1.480	
M3bcRQ3	Cat-	fish	biomass	svs_ref- er- ence	full Substrate_in- dex_after	0.571	1.054	0.541	0.588	NA	-	2.637
											1.495	
M3bcRQ3	Cat-	fish	biomass	svs_ref- er- ence	full site- Stone_pile:mon- itoring_int	1.223	0.642	1.905	0.057	0.114	-	2.481
											0.035	
M4TBQ4	Kōura	abun-	treat-	ment	be- (Intercept)	-	0.785	-	0.016	NA	-	-
		dance				1.897		2.415			3.437	0.358
M4TBQ4	Kōura	abun-	treat-	ment	be- Site_type_de-	0.288	1.044	0.276	0.783	0.783	-	2.334
		dance			tailStone_pile						1.759	
M4TBQ4	Kōura	biomass	treat-	ment	be- (Intercept)	1.030	0.830	1.241	0.215	NA	-	2.656
											0.597	
M4TBQ4	Kōura	biomass	treat-	ment	be- Site_type_de-	1.044	0.977	1.068	0.286	0.571	-	2.959
					tailStone_pile						0.872	
M4TBQ4	Cat-	fish	abun-	treat-	be- (Intercept)	-	0.908	-	0.011	NA	-	-
		dance				2.303		2.535			4.083	0.522
M4TBQ4	Cat-	fish	abun-	treat-	be- Site_type_de-	-	1.535	-	0.652	0.783	-	2.315
		dance			tailStone_pile	0.693		0.452			3.701	
M4TBQ4	Cat-	fish	biomass	treat-	be- (Intercept)	3.203	1.279	2.504	0.012	NA	0.696	5.709
M4TBQ4	Cat-	fish	biomass	treat-	be- Site_type_de-	-	3.107	-	0.101	0.403	-	0.990
					tailStone_pile	5.100		1.641			11.190	

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model	Q1	species	response	com- pari- type	time_frame	term	es- ti- mate	std. error	statis- tic	p.value	adjco- B	low	high
M4RBBQ4	Kōura	abun-	ref-	be-	(Intercept)		0.742	0.553	1.343	0.179	NA	-	1.825
		dance	er-	fore								0.341	
			ence										
M4RBBQ4	Kōura	abun-	ref-	be-	Site_type_de-		0.031	0.778	0.040	0.968	0.968	-	1.557
		dance	er-	fore	tailRefer-							1.494	
			ence		ence_2								
M4RBBQ4	Kōura	biomass	ref-	be-	(Intercept)		3.623	0.478	7.580	0.000	NA	2.686	4.560
			er-	fore									
			ence										
M4RBBQ4	Kōura	biomass	ref-	be-	Site_type_de-		-	0.681	-	0.952	0.968	-	1.293
			er-	fore	tailRefer-		0.041		0.061			1.375	
			ence		ence_2								
M4RBBQ4	Cat-	abun-	ref-	be-	(Intercept)		-	0.715	-	0.142	NA	-	0.353
	fish	dance	er-	fore			1.050		1.467			2.452	
			ence										
M4RBBQ4	Cat-	abun-	ref-	be-	Site_type_de-		-	1.074	-	0.754	0.968	-	1.769
	fish	dance	er-	fore	tailRefer-		0.336		0.313			2.442	
			ence		ence_2								
M4RBBQ4	Cat-	biomass	ref-	be-	(Intercept)		4.989	1.054	4.731	0.000	NA	2.922	7.055
	fish		er-	fore									
			ence										
M4RBBQ4	Cat-	biomass	ref-	be-	Site_type_de-		-	1.634	-	0.493	0.968	-	2.082
	fish		er-	fore	tailRefer-		1.122		0.686			4.325	
			ence		ence_2								
M4TABQ4	Kōura	abun-	treat-	af-	(Intercept)		0.772	1.071	0.720	0.471	NA	-	2.871
		dance	ment	ter								1.328	
M4TABQ4	Kōura	abun-	treat-	af-	Site_type_de-		0.054	0.795	0.067	0.946	NA	-	1.612
		dance	ment	ter	tailStone_pile							1.505	

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model	re- Q	com- pari- response_type	time_frame	es- ti- std.er-stat- tic	p.value	adjco	BH	low	high
M4TABQ4	Kōura	abun-	treat-	af-	monitoring_int	- 0.165	- 0.069	NA	- 0.023
		dance	ment	ter		0.299	1.820	0.622	
M4TABQ4	Kōura	abun-	treat-	af-	Temperature	- 0.044	- 0.220	NA	- 0.032
		dance	ment	ter		0.053	1.226	0.139	
M4TABQ4	Kōura	abun-	treat-	af-	Site_type_de-	0.328	0.203	1.619	0.105
		dance	ment	ter	tail-			0.112	- 0.726
					Stone_pile:mon-			0.069	
					itoring_int				
M4TABQ4	Kōura	biomas	treat-	af-	(Intercept)	3.154	0.872	3.618	0.000
		ment	ter					NA	1.445
								4.863	
M4TABQ4	Kōura	biomas	treat-	af-	Site_type_de-	- 0.996	- 0.704	NA	- 1.574
		ment	ter		tailStone_pile	0.378	0.379	2.329	
M4TABQ4	Kōura	biomas	treat-	af-	monitoring_int	- 0.220	- 0.117	NA	- 0.086
		ment	ter			0.345	1.568	0.775	
M4TABQ4	Kōura	biomas	treat-	af-	Site_type_de-	0.459	0.276	1.661	0.097
		ment	ter		tail-			0.112	- 1.000
					Stone_pile:mon-			0.083	
					itoring_int				
M4TABQ4	Cat-	abun-	treat-	af-	(Intercept)	- 1.629	- 0.169	NA	- 0.950
	fish	dance	ment	ter		2.243	1.377	5.435	
M4TABQ4	Cat-	abun-	treat-	af-	Site_type_de-	- 1.839	- 0.150	NA	- 0.959
	fish	dance	ment	ter	tailStone_pile	2.645	1.439	6.248	
M4TABQ4	Cat-	abun-	treat-	af-	monitoring_int	- 0.458	- 0.201	NA	- 0.313
	fish	dance	ment	ter		0.586	1.278	1.484	
M4TABQ4	Cat-	abun-	treat-	af-	Site_type_de-	1.166	0.586	1.990	0.047
	fish	dance	ment	ter	tail-			0.112	0.018
					Stone_pile:mon-			2.315	
					itoring_int				

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q1	species	response	type	com- pari- time	frame	es- ti- mate	std.er	statis- tic	p.value	adjco- B	How	conf.high
M4TABQ4	Cat-	abun-	treat-	af-	(Intercept)		-	19359.257	-	0.999	NA	-	37923.044
	fish	dance	ment	ter			20.402	0.001			37963.849		
M4TABQ4	Cat-	biomas-	treat-	af-	(Intercept)		3.571	2.008	1.778	0.075	NA	-	7.508
	fish		ment	ter							0.365		
M4TABQ4	Cat-	biomas-	treat-	af-	Site_type_de-		-	4.577	-	0.160	NA	-	2.539
	fish		ment	ter	tailStone_pile		6.432	1.405			15.402		
M4TABQ4	Cat-	biomas-	treat-	af-	monitoring_int		-	0.688	-	0.209	NA	-	0.484
	fish		ment	ter			0.864	1.256			2.212		
M4TABQ4	Cat-	biomas-	treat-	af-	Site_type_de-		1.928	1.214	1.588	0.112	0.112	-	4.307
	fish		ment	ter	tail-						0.451		
					Stone_pile:mon-								
					itoring_int								
M4RABQ4	Kōura	abun-	ref-	af-	(Intercept)		-	0.976	-	0.448	NA	-	1.173
		dance	er-	ter			0.740	0.758			2.652		
			ence										
M4RABQ4	Kōura	abun-	ref-	af-	Site_type_de-		-	0.601	-	0.710	NA	-	0.955
		dance	er-	ter	tailRefer-		0.223	0.372			1.402		
			ence		ence_2								
M4RABQ4	Kōura	abun-	ref-	af-	monitoring_int		-	0.094	-	0.072	NA	-	0.015
		dance	er-	ter			0.169	1.801			0.353		
			ence										
M4RABQ4	Kōura	abun-	ref-	af-	Temperature		0.071	0.034	2.091	0.037	NA	0.004	0.138
		dance	er-	ter									
			ence										
M4RABQ4	Kōura	abun-	ref-	af-	Site_type_de-		-	0.124	-	0.761	0.902	-	0.206
		dance	er-	ter	tailRefer-		0.038	0.304			0.281		
			ence		ence_2:moni-								
					toring_int								

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	re-species	com- pari- response_type	time_frame	term	es- mate	std.er-	statis- tic	p.value	adjco- B	How	low	high
M4RA04	Kōura	biomass	ref- er- ence	af- ter	(Intercept)	3.370	0.860	3.917	0.000	NA	1.683	5.056
M4RA04	Kōura	biomass	ref- er- ence	af- ter	Site_type_de- tailRefer- ence_2	- 0.184	0.737 0.250	- 0.803	NA	- 1.630	-	1.261
M4RA04	Kōura	biomass	ref- er- ence	af- ter	monitoring_int	- 0.104	0.117 0.891	- 0.373	NA	- 0.334	-	0.125
M4RA04	Kōura	biomass	ref- er- ence	af- ter	Site_type_de- tailRefer- ence_2:moni- toring_int	- 0.040	0.167 0.237	- 0.812	0.902	- 0.368	-	0.288
M4RA04	Cat- fish	abun- dance	ref- er- ence	af- ter	(Intercept)	- 6.715	1.788 3.757	- 0.000	NA	- 10.219	-	3.212
M4RA04	Cat- fish	abun- dance	ref- er- ence	af- ter	Site_type_de- tailRefer- ence_2	0.731	1.033 0.707	0.479	NA	- 1.295	-	2.756
M4RA04	Cat- fish	abun- dance	ref- er- ence	af- ter	monitoring_int	- 0.051	0.233 0.220	- 0.826	NA	- 0.509	-	0.406
M4RA04	Cat- fish	abun- dance	ref- er- ence	af- ter	Temperature	0.258	0.082	3.131	0.002	NA	0.096	0.419

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Re- Q1	species	response	com- pari- type	time	frame	es- ti- mate	std.er- ror	statis- tic	p.value	adjco- B	low	high
M4RAQ4	Cat-	abun-	ref-	af-	Site_type_de-	-	0.296	-	0.623	0.902	-	0.435	
	fish	dance	er-	ter	tailRefer-	0.146		0.491			0.727		
			ence		ence_2:moni-								
					toring_int								
M4RAQ4	Cat-	abun-	ref-	af-	(Intercept)	-	13711.191	-	0.999	NA	-	26853.421	
	fish	dance	er-	ter		20.018		0.001			26893.458		
			ence										
M4RAQ4	Cat-	biomass	ref-	af-	(Intercept)	2.128	1.527	1.393	0.163	NA	-	5.121	
	fish		er-	ter							0.865		
			ence										
M4RAQ4	Cat-	biomass	ref-	af-	Site_type_de-	0.062	1.575	0.039	0.969	NA	-	3.149	
	fish		er-	ter	tailRefer-						3.025		
			ence		ence_2								
M4RAQ4	Cat-	biomass	ref-	af-	monitoring_int	0.007	0.325	0.023	0.982	NA	-	0.644	
	fish		er-	ter							0.629		
			ence										
M4RAQ4	Cat-	biomass	ref-	af-	Site_type_de-	0.055	0.448	0.123	0.902	0.902	-	0.933	
	fish		er-	ter	tailRefer-						0.823		
			ence		ence_2:moni-								
					toring_int								

Table 3: DHARMA goodness-of-fit tests per model. $p < 0.05$ indicates a significant departure from model assumptions (uniformity of residuals, over/under-dispersion, or excess outliers) and warrants closer inspection of the corresponding diagnostic plot above.

Model	Uniformity_p	Dispersion_p	Outlier_p
M1ok	0.115	0.972	1
M1ak	0.791	0.912	1
M1bk	0.453	0.810	1
M1oc	0.504	0.700	1
M1bc	0.546	0.406	1
M2ak	0.326	0.736	1
M2bk	0.633	0.720	1
M2ac	0.481	0.208	1
M2bc	0.400	0.186	1
M3ak	0.856	0.626	1
M3bk	0.972	0.602	1
M3ac	0.473	0.374	1
M3bc	0.580	0.260	1
M4TAak	0.416	0.970	1
M4TAbk	0.825	0.866	1
M4TAbc	0.697	0.668	1
M4RAak	0.956	0.874	1
M4RAbk	0.872	0.896	1
M4RAbc	0.885	0.748	1

20 Source: [Article Notebook](#)

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